

SCHOLAR: Inspectable Local-First Question Answering for Scientific Papers

Prithviraj Patil

Independent Researcher

prithvisp28@gmail.com

Abstract

Researchers often need to verify a paper claim against the exact paragraph, table, or figure that supports it. SCHOLAR is a local-first scientific-document assistant built around that interaction: a user asks a question, receives a cited answer, and clicks a citation to inspect the linked source region in the original PDF. The system preserves page identity and layout coordinates during ingestion, combines lexical and dense retrieval, and records the evidence and verifier actions behind each response. In a retrospective benchmark of 100 questions across 25 papers, fused retrieval reaches gold-page Hit@1 of 0.85; a separate 150-case language-model-based audit reports grounding and citation categories, but is not human ground truth. The demonstration shows how researchers, students, and reviewers can use this inspectable workflow locally, with the final submission providing an installable package and screencast link.

1 Introduction

Researchers, students, and peer reviewers use document assistants to extract experimental results, definitions, and visual evidence from scientific PDFs. These documents are unusually difficult for question answering because a claim may depend on text, a table, a figure, or the spatial relationship between them. Existing retrieval-augmented generation (RAG) systems (Lewis et al., 2020) can produce plausible answers, but following one citation back to the exact supporting region is still awkward, especially when an unpublished paper should remain on the reader’s machine.

We demonstrate SCHOLAR as a concrete interaction: upload a paper, ask a question, receive a cited answer, click a citation, and inspect the corresponding paragraph, figure, or table region in the original PDF. The system keeps the evidence object—not merely a text snippet—attached to the

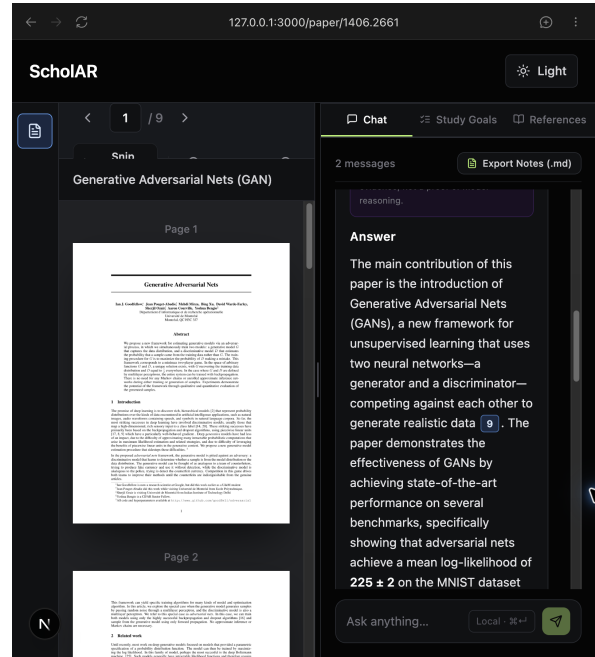


Figure 1: **The interaction the demo exposes.** The workspace places the original PDF beside a cited answer; citation chips are the entry point to page and region inspection. The screencast shows the synchronized jump and evidence-inspection path.

answer and records retrieval, citation, verification, and final-output decisions in an audit trace.

Scientific-document understanding benchmarks such as QASPER, SciFact, and M3SciQA show the need to retrieve and reason over scholarly evidence (Dasigi et al., 2021; Wadden et al., 2020; Li et al., 2024); citation-grounded generation studies examine whether retrieved context supports generated claims (Gao et al., 2023; Asai et al., 2024); and multimodal document datasets show why page layout, figures, and tables matter (Pramanick et al., 2024; Faysse et al., 2025). Related EACL demonstrations address RAG evaluation and historical-manuscript analysis (Es et al., 2024; Sakhovskiy et al., 2026). SCHOLAR differs in combining local execution with citations that retain spatial identity

and can be inspected in the original PDF.

In this demonstration, we make three contributions:

1. **Inspectable scientific QA:** A local-first assistant whose answer citations resolve to text bounding boxes, figures, and tables in the original PDF.
2. **Auditable grounding pipeline:** Layout-aware document representation, hybrid retrieval, and deterministic lexical/numerical checks preserve the connection between generated claims and retrieved evidence.
3. **Open interactive system:** A reproducible implementation characterized with retrieval, grounding, and consumer-workstation diagnostics, with the package and screencast supplied as submission artifacts.

The software is MIT-licensed; access details are given in Section 6.

2 System Architecture

SCHOLAR is a local interactive system: a Python/FastAPI backend prepares source-scoped evidence and a Next.js/React frontend presents the answer beside the document. Figure 2 summarizes the three stages that matter to a reader: document representation, evidence retrieval and local answering, and verification and inspection.

2.1 Document representation

When a user provides a scientific PDF, the ingestion engine extracts both structural semantics and visual coordinates:

- A layout-aware parser recovers document hierarchy while isolating figures and tables, with a PyMuPDF fallback; the primary parser is Docling (Auer et al., 2024).
- Each text or visual element receives a source-scoped identity containing its document, page, and normalized bounding box.
- Figures, tables, and page regions remain available as local crops; ingestion commits atomically so a partial paper is not queryable.

2.2 Evidence retrieval and local answering

BM25 (Robertson and Zaragoza, 2009) and dense text retrieval are fused with reciprocal-rank fusion (Cormack et al., 2009); visual crops and full

pages provide candidates when a question refers to a figure or table. The application—not the model—owns page numbers and bounding boxes. Retrieved evidence is placed in a structured prompt for a local Ollama model such as qwen3.5:9b (Qwen Team, 2026); the model emits citation markers referring to evidence items, while the application maps those markers back to source identities. Strict-local mode blocks non-loopback network access during analysis, separating local answering from any explicitly enabled acquisition step.

2.3 Verification and inspection

After generation, SCHOLAR runs a post-generation inspection stage via `verifier.service`:

- lexical overlap and numerical checks flag mismatches between cited claims and supplied evidence;
- bounded remapping or pruning records changes without generating replacement facts; and
- the query, retrieved objects, raw completion, verifier actions, final citations, and timings are stored in a versioned trace that the Evidence Graph view exposes.

The verifier is a routing and inspection aid, not semantic entailment or human-verified ground truth. The user’s central action remains direct inspection of the original PDF region.

3 Interactive Demonstration

The core user experience of SCHOLAR is direct verification: readers can check the evidence behind each citation in the same workspace as the answer. Figure 1 shows the workspace; the screencast supplies the full click-to-region sequence.

3.1 Representative scenarios

The demonstration uses three short scenarios.

Text and numerical verification. The user uploads a paper and asks, “What improvement does the proposed method report over baseline X?” SCHOLAR retrieves the relevant passages and tables, produces a cited answer, and lets the user click the citation to inspect the source page and region.

Visual evidence inspection. The user asks, “What does Figure 3 show about the ablation?” SCHOLAR supplies the figure or table crop with its caption and keeps the citation linked to the original

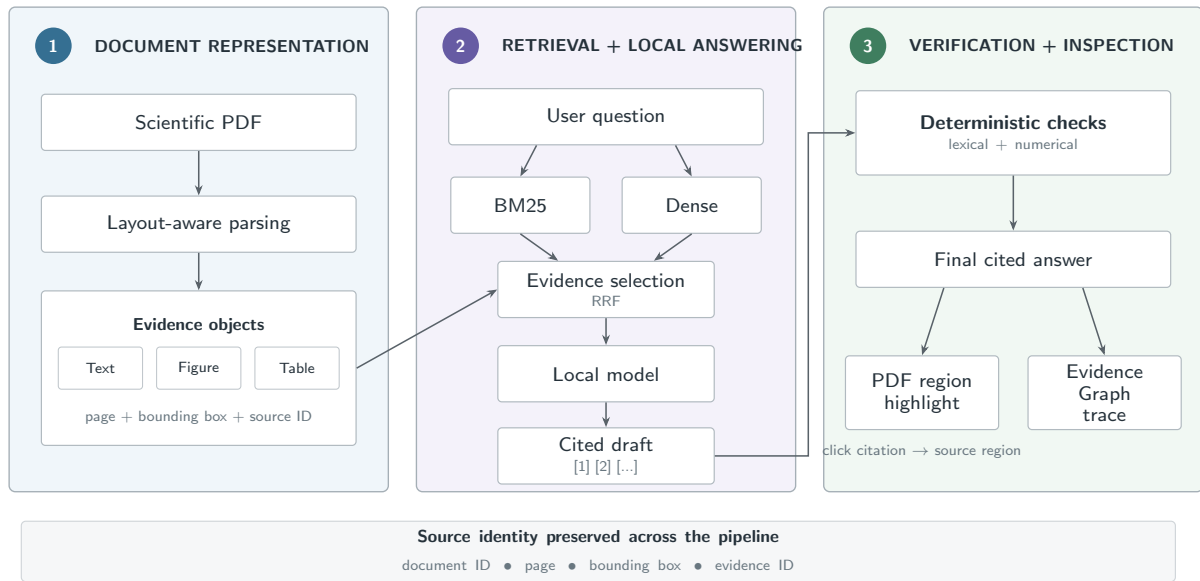


Figure 2: **SCHOLAR architecture.** A scientific PDF is converted into source-linked evidence objects, which support lexical and dense retrieval for local cited answering. Deterministic checks produce the final answer while preserving links to inspectable PDF regions and an auditable execution trace.

page, allowing the reader to compare the generated description with the source pixels.

Insufficient evidence and audit. The user asks for a fact that is not present in the retrieved paper. The system can expose weak or missing support rather than presenting provenance as proof; the user then opens the trace or Evidence Graph to inspect retrieved candidates, citation origins, verifier actions, and terminal status.

The live demonstration follows the complete path: upload and index a PDF, ask one text or visual question, inspect the cited region, open the graph, and export the trace. Clicking a citation synchronizes the answer and viewer; the interface distinguishes text evidence from figure/table crops and displays the current verification status.

3.2 Intended audience and interface

SCHOLAR targets academic researchers comparing results, peer reviewers checking claims, and graduate students reading technical papers. The value of the demo is not a claim that every answer is correct; it is that the user can move from generated prose to the original evidence with a visible, inspectable path.

Clicking a citation marker triggers an immediate synchronization event:

- the document viewer navigates to the cited page and evidence region;

- the selected text, figure, or table remains distinguishable from the generated answer; and
- the trace and Evidence Graph expose retrieved candidates, citation origins, verifier actions, and final status.

4 Execution Contract and Traceability

The interaction is governed by a structured execution contract rather than an unconstrained text-generation call. For each question, the backend records the input document and query, ranked evidence identities and retrieval scores, the exact evidence supplied to the generator, the raw completion, verifier actions, final citations, and stage timings. This record is persisted as a validated JSON trace under `backend/data/traces/`; the Evidence Graph view exposes the same provenance to the reader.

4.1 Citation-constrained generation

The local model receives a numbered set of evidence excerpts whose identifiers are assigned by the application. The system prompt instructs it to answer only from those excerpts, attach a citation marker to every factual or numerical claim, and abstain when the excerpts are insufficient. A shortened version of the prompt is:

```
Answer strictly using
the provided evidence
excerpts. Every factual
```

claim, quantitative value, or mathematical statement must be followed by one or more markers such as [1] or [1, 2]. Do not introduce facts absent from the excerpts; if the evidence is insufficient, say so explicitly.

The model therefore proposes citation markers, but it does not own page numbers or bounding boxes. The application resolves each marker to the source-scoped evidence identity, preserves the original page and normalized coordinates, and retains the unmodified completion for auditability.

4.2 Trace fields and inspection path

The trace schema makes the provenance path inspectable without treating it as proof of semantic entailment. Its principal fields are:

- `query_id`, `paper_id`, and `raw_query` identify the transaction and source document;
- `retrieved_chunks` stores ranked chunk IDs, fusion scores, page numbers, modalities, and normalized boxes;
- `raw_completion` preserves the exact local-model output before repair;
- `verifier_actions` records citation remapping, pruning, and abstention decisions; and
- `final_answer` and `latency_breakdown` record the user-visible result and timing by stage.

Clicking a citation uses these application-owned identities to select the corresponding page and region in the PDF viewer. The same identity is available to the Evidence Graph, so a reader can move from the answer to the retrieved candidate, verifier decision, and original source region.

5 Evaluation

5.1 Can the system find relevant evidence?

We evaluate page-level evidence retrieval on a retrospective benchmark of 100 questions across 25 scientific papers. Table 1 compares lexical, dense, and fused retrieval configurations. BM25 places a gold evidence page at rank 1 for 84% of queries (MRR 0.892). BM25+dense reciprocal-rank fusion reaches Hit@1 of 0.85 and MRR of 0.905, but its paired bootstrap Hit@1 difference over BM25 is only +0.01 (95% CI $[-0.054, +0.078]$), which includes zero. An uncalibrated modality boost instead drops Hit@1 from 0.85 to 0.59 (-0.26 , 95% CI $[-0.388, -0.136]$). We therefore keep visual

Retrieval Configuration	Hit@1 $\uparrow$	Hit@5 $\uparrow$	MRR $\uparrow$	nDCG@5 $\uparrow$
Keyword overlap	0.71	0.91	0.796	0.757
BM25 lexical	0.84	0.97	0.892	0.855
Dense embedding	0.76	0.95	0.838	0.796
BM25 + dense RRF	0.85	0.98	0.905	0.853
BM25 + dense + modality RRF	0.59	0.95	0.734	0.714

Table 1: **Retrospective Page-Retrieval Ablation** on 100 questions across 25 scientific papers. Fusing BM25 and dense retrieval with Reciprocal Rank Fusion (RRF) yields Hit@1 of 0.85 (paired bootstrap difference +0.01, 95% CI $[-0.054, +0.078]$ over BM25). Adding an uncalibrated modality heuristic degrades Hit@1 to 0.59 (-0.26 , 95% CI $[-0.388, -0.136]$).

evidence available for inspection without treating visual intent as proof that a candidate is relevant.

5.2 Human evaluation

The human study uses the same 50 questions drawn from 10 scientific papers, with five independent evaluators and 250 planned ratings. The stratified sample contains 10 direct text lookups, 23 multi-passage synthesis questions, 8 figure/visual-evidence questions, 3 mathematical-mechanism questions, and 6 table/quantitative questions; difficulty is split into 10 easy, 20 medium, and 20 hard cases (Table 3).

Status. The protocol is active and the ratings are being collected; verified aggregates are intentionally withheld from this working PDF. This gate prevents provisional or fabricated values from appearing in a submission. Once the ratings are frozen, replace the guarded macros in `build/evaluation_gates.tex`, set the results gate to true, and recompile.

5.3 Secondary automated audit and repair diagnostic

The completed 150-case LLM-based evaluation provides a categorical diagnostic of answer correctness, evidence grounding, and citation quality. The retained rows identify the evaluator generically, so we do not claim a named judge model; visual labels are excluded because the later visual update introduced a value outside the frozen rubric. On the 113 reference-eligible cases, 59 are correct, 33 partly correct, 20 incorrect, and one not judgeable. Across all 150 cases, grounding is supported in 112 and citation quality is accurate in 110. These are model judgments, not human ground truth.

The deterministic repair ablation remains informative but not a headline quality result: it raises reference-unit overlap F1 from 0.159 to 0.224

Dimension	Supported/Correct	Partial	Incorrect/Unsupp.	N/A
Correctness (eligible $n = 113$)	59 (52.2%)	33 (29.2%)	20 (17.7%)	1 (0.9% unjudgeable)
Completeness ($n = 150$)	83 (55.3%)	24 (16.0%)	41 (27.3%)	2 (not judgeable)
Grounding ($n = 150$)	112 (74.7%)	26 (17.3%)	10 (6.7%)	2 (not judgeable)
Citation Quality ($n = 150$)	110 (73.3%)	28 (18.7%)	10 (6.7%)	2 (missing)

Table 2: **LLM-based evaluation results on 150 complement cases.** Correctness is evaluated strictly against the 113 verified eligible references (*37 cases had ineligible reference answers; 1 eligible case was unjudgeable). Grounding and citation quality evaluate local excerpt attribution. The table reports only categories in the frozen rubric; a later visual-label update introduced an out-of-schema category and is excluded. These are model judgments, not human ratings.

while lowering cited-page alignment from 0.670 to 0.618 on the 20 gold-page cases. We therefore treat repair as a diagnostic and preserve the principle that spatially correct evidence matters more than lexical overlap alone; the paired diagnostic is reported in Table 4.

5.4 Can the complete path run locally?

Across 100 valid requests on an Apple M3 Pro with 18 GB unified memory and local Ollama, end-to-end latency has $p_{50} = 27.34$ s, $p_{90} = 41.61$ s, and $p_{95} = 149.54$ s. Retrieval has median 523 ms; generation has median 26.59 s and $p_{95} = 37.48$ s; verification and trace writing have median 15 ms and $p_{95} = 24$ ms. These observations establish feasibility on the named workstation, not interactive responsiveness at scale.

Taken together, these results support a narrow claim: the current prototype retrieves useful evidence and runs locally on the tested workstation. They do not establish generalization or human benefit.

6 Availability and Reproducibility

Access. SCHOLAR is MIT-licensed; its source is at <https://github.com/prithvi-kaizen/Scholar>. The final submission will add a signed-out-accessible, versioned ZIP URL with a SHA-256 checksum and a screencast URL (no longer than 2.5 minutes). The release contains a self-authored example PDF; private papers, caches, traces, and model weights are excluded.

Tested environment. The verified profile uses Python 3.12, Node.js 20/npm 10, Ollama, and a Q4_K_M qwen3.5:9b model on an Apple M3 Pro with 18 GB unified memory.

7 Conclusion

We presented SCHOLAR, a local-first scientific-paper assistant built around one interaction: ask a question, then inspect the exact source region attached to each citation. Layout-aware evidence objects, hybrid retrieval, deterministic checks, and an execution trace support that interaction without turning provenance into a claim of factual entailment. The retrospective diagnostics show useful retrieval and feasible local operation, while also showing why lexical repair and model judgments require caution. The demo’s contribution is an inspectable workflow for researchers, reviewers, and students; its package and screencast must make that workflow directly testable.

Limitations

The current system and evidence have four clear limits.

Layout and parsing. The dual-engine ingestion pipeline works best on modern two-column preprints. Degraded scans, multi-page tables, intricate vector graphics, and formulas that depend on custom fonts can produce misaligned boxes or incomplete text and table extraction.

Verifier scope. The verifier checks lexical overlap, n-gram containment, and numerical patterns. It does not establish semantic entailment or prove a mathematical statement. It can therefore miss a conceptual error that reuses source vocabulary, or remove a valid paraphrase whose wording differs substantially from the source.

Hardware and latency. The measured profile covers one Apple M3 Pro workstation with 18 GB unified memory. End-to-end latency is $p_{50} = 27.34$ s and $p_{95} = 149.54$ s under uncontrolled cache conditions. Other devices may be slower or may not fit the local model; we make no claim about responsiveness at scale.

Evidence scope. The retrieval benchmark contains 100 retrospective questions over 25 papers, so its results are not population estimates. The 150-case language-model audit is not human ground truth, and no real researcher workflow or user benefit has yet been measured.

8 Reproducibility details

The human-study sampling frame is stratified by question type and difficulty. Table 3 records the planned 50-question sample; five independent evaluators rate the same questions, for 250 planned ratings. Human aggregates remain gated until the ratings and quality checks are complete.

Question Category	Easy	Medium	Hard	Total
Direct text lookup	1	5	4	10
Multi-passage synthesis	4	10	9	23
Figure / visual evidence	2	2	4	8
Mathematical mechanism	1	1	1	3
Table / quantitative	2	2	2	6
Total Stratified Sample	10	20	20	50

Table 3: **Planned human-evaluation sample.** The same 50 questions are rated independently by five evaluators across 10 scientific papers ($n = 250$ ratings planned).

The audited local profile uses an Apple M3 Pro with 18 GB unified memory, Python 3.12, Node.js 20/npm 10, Ollama v0.5.12, and a Q4_K_M qwen3.5:9b model. The measured 100-request profile establishes feasibility on this workstation; it is not a throughput or responsiveness claim.

For completeness, Table 4 reports the paired di-

agnostic for the post-generation repair step. We keep it separate from the headline retrieval result because the repair improves lexical overlap while reducing citation-page alignment on the gold-page subset.

Automatic Diagnostic	Off	On	Paired Change (95% CI)
Reference-unit overlap F1	0.159	0.224	+0.065 [+0.042, +0.091]
Automatic key-term coverage	0.759	0.742	-0.017 [-0.038, 0.000]
Citation-page alignment ($n = 20$)	0.670	0.618	-0.052 [-0.091, -0.015]
Pipeline SUCCESS status	28/30	30/30	+2/30

Table 4: **Retrospective Diagnostic of Post-Generation Repair** on 30 paired questions across 19 papers with identical raw completions. Repair raises lexical overlap F1 but reduces cited-page alignment on gold-page cases. Overlap measures lexical surface matching, not factual correctness.

Artifact smoke test. The clean-environment smoke-test log exercises layout-aware ingestion with its fallback path, strict-local network policy, fused BM25+dense retrieval, deterministic citation repair, and the offline human-evaluation schema. The recorded checks pass without bundling private papers, model caches, raw traces, or provisional human outcomes; the same boundary is used for the submission package.

Ethical Considerations

Privacy and local execution. SCHOLAR is intended for unpublished manuscripts, review drafts, and proprietary notes. In strict-local analysis mode, the network policy blocks non-loopback requests. Users must still distinguish that analysis boundary from any explicitly enabled document, model, or reference acquisition.

Copyright and redistribution. The demonstration package and public repositories do not redistribute copyrighted publications without authorization. Benchmarks and tutorial examples use open-access preprints distributed under Creative Commons or arXiv non-exclusive licenses.

Evaluators and consent. The five-rater offline study is configured for collection over the same 50 questions and 10 papers described in Table 3. Evaluators provide informed consent and use pseudonymous identifiers (e.g., R01–R05); public aggregates will not expose contact details or identifiable comments. Results are not included until the ratings and quality checks are complete.

External models. Secondary evaluation with an external model requires explicit operator authorization and sets `store=false`. Users are told that external APIs are not zero-retention environments.

References

- Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2024. [Self-RAG: Learning to retrieve, generate, and critique through self-reflection](#). In *The Twelfth International Conference on Learning Representations*.
- Christoph Auer, Maksym Lysak, Ahmed Nassar, Michele Dolfi, Nikolaos Livathinos, Panos Vagenas, Cesar Berrospi Ramis, Matteo Omenetti, Fabian Lindlbauer, Kasper Dinkla, Lokesh Mishra, Yusik Kim, Shubham Gupta, Rafael Teixeira de Lima, Valery Weber, Lucas Morin, Ingmar Meijer, Viktor Kuropiatnyk, and Peter W. J. Staar. 2024. [Docling technical report](#). Technical Report arXiv:2408.09869, arXiv.
- Gordon V. Cormack, Charles L. A. Clarke, and Stefan Buettcher. 2009. [Reciprocal rank fusion outperforms condorcet and individual rank learning methods](#). In *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 758–759.
- Pradeep Dasigi, Kyle Lo, Iz Beltagy, Arman Cohan, Noah A. Smith, and Matt Gardner. 2021. [A dataset of information-seeking questions and answers anchored in research papers](#). In *Proceedings of NAACL-HLT*, pages 4599–4610.
- Shahul Es, Jithin James, Luis Espinosa-Anke, and Steven Schockaert. 2024. [RAGAs: Automated evaluation of retrieval augmented generation](#). In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*, pages 150–158.
- Manuel Faysse, Hugues Sibille, Tony Wu, Bilel Omrani, Gautier Viaud, Céline Hudelot, and Pierre Colombo. 2025. [ColPali: Efficient document retrieval with vision language models](#). In *International Conference on Learning Representations*.
- Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. 2023. [Enabling large language models to generate text with citations](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 6465–6488.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich K’uttler, Mike Lewis, Wen-tau Yih, Tim Rockt’aschel, Sebastian Riedel, and Douwe Kiela. 2020. [Retrieval-augmented generation for knowledge-intensive NLP tasks](#). In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474.
- Chuhan Li, Ziyao Shanguan, Yilun Zhao, Deyuan Li, Yixin Liu, and Arman Cohan. 2024. [M3SciQA: A multi-modal multi-document scientific QA benchmark for evaluating foundation models](#). In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 15419–15446.
- Shraman Pramanick, Rama Chellappa, and Subhashini Venugopalan. 2024. [SPIQA: A dataset for multi-modal question answering on scientific papers](#). In *Advances in Neural Information Processing Systems 37, Datasets and Benchmarks Track*, volume 37.
- Qwen Team. 2026. [Qwen3.5-9B model card](#). Hugging Face model repository.
- Stephen Robertson and Hugo Zaragoza. 2009. [The probabilistic relevance framework: BM25 and beyond](#). *Foundations and Trends in Information Retrieval*, 3(4):333–389.
- Andrey Sakhovskiy, Ivan Ulitin, Emilia Bojarskaja, Vladimir Kokh, Ruslan Murtazin, Maxim Novopol’tsev, and Semen Budenny. 2026. [InkSight: Towards AI-aided historical manuscript analysis](#). In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 3: System Demonstrations)*, pages 271–281.
- David Wadden, Shanchuan Lin, Kyle Lo, Lucy Lu Wang, Madeleine van Zuylen, Arman Cohan, and Hannaneh Hajishirzi. 2020. [Fact or fiction: Verifying scientific claims](#). In *Proceedings of EMNLP*, pages 7534–7550.

A Supplementary Audit Artifacts

The release preserves the machine-readable evaluation tables, the guarded human-evaluation macros, and the JSON trace schema used by the demonstration. These supplementary artifacts support reproduction without adding provisional human outcomes to the paper.

A.1 Execution-trace schema

Each completed query is serialized as a validated JSON trace. The fields below are the stable inspection boundary exposed by the backend and Evidence Graph; values are recorded per request rather than aggregated into a new evaluation result.

Field	Recorded information
query_id, paper_id raw_query	Transaction and source-document identities. User question supplied to the pipeline.
retrieved_chunks	Ranked evidence IDs, fusion scores, page numbers, modalities, and normalized boxes.
raw_completion	Unmodified local-model completion before repair.
verifier_actions	Citation remapping, pruning, and abstention decisions.
final_answer	User-visible answer and retained citation markers.
latency_breakdown	Timing recorded for each pipeline stage.

Table 5: Stable fields in the inspectable execution trace. Page and bounding-box identity is carried by the retrieved evidence objects, not generated by the local model.

A.2 Generation contract

The generator receives a numbered set of application-owned evidence excerpts. The release prompt instructs the model to answer only from those excerpts, attach one or more markers (for example, [1] or [1, 2]) to each factual, numerical, or mathematical claim, and state explicitly when the supplied evidence is insufficient. The model proposes markers but never assigns page numbers or coordinates; the application resolves markers to the source-scoped evidence identities retained in the trace.

A.3 Reproduction checklist

The installable package is intended to make the demonstration inspectable in four steps: (i) start the local backend and frontend, (ii) load the self-authored example PDF included with the release, (iii) ask a text or visual question and open a citation, and (iv) export the JSON trace and inspect the linked page region. The tested local profile is documented in Section 8; private papers, caches, traces, and model weights are not bundled.

A.4 Evaluation-status boundary

The machine-readable retrieval, language-model audit, repair diagnostic, and hardware-profile artifacts are included for reproducibility. The human-evaluation gate remains closed until the planned ratings and quality checks are complete; this appendix therefore contains no provisional human aggregates.